HYDROFARMS AGRI SOLUTIONS

# GROUNDWATER TREATMENT, REVERSE OSMOSIS, UV DISINFECTION AND TREATED-WATER TRANSFER

## Activity-Specific Construction Method Statement

| Document field | Controlled value |
|---|---|
| Document code | HF-GH01-MST-WTR-001 |
| Revision | 00 - Draft for coordination |
| Project | One-Feddan NFT Lettuce Greenhouse Project |
| Location | Alexandria, Egypt |
| Source and duty | Groundwater well; 5.0 m3/h net treated-water output |
| Prepared by | Hydrofarms Agri Solutions |
| Issue date | 15 August 2026 |
| Related method | HF-GH01-MST-NFT-001 Nutrient Circulation System |

**TREATED WATER ONLY:** The reserve group and germination tank receive verified treated water. Raw water, backwash, concentrate and CIP waste have no cross-connection to treated storage.

# Document Control

## Approval and responsibility

| Role | Primary responsibility |
|---|---|
| Project manager | Resources, programme, supplier/subcontractor control and procurement status. |
| Construction manager | Work-front release, interfaces, sequencing and service coordination. |
| Water-treatment engineer | Water analysis, process projection, equipment sizing and performance guarantee. |
| Hydraulic engineer | Routes, sizing, valves, tank interface and pressure tests. |
| Electrical/controls engineer | Power, PLC, instruments, alarms and cause-and-effect. |
| QA/QC and HSE | Traceability, tests, chemicals, pressure, wet electrical work and waste control. |
| Client/consultant | Hold/witness inspection and technical acceptance. |

# 1. Purpose

This method controls procurement, installation, flushing, testing, commissioning and handover of the groundwater treatment and reverse-osmosis package from the well interface through pretreatment, RO and UV to treated-water storage.

# 2. Scope and Package Boundary

- 3 in raw-water line and final 2 in WTP interface with booster pump.
- Complete pretreatment, conditioning, RO, UV, instrumentation and PLC package.
- 2 in product route, reserve fill header, four 1.5 in branches and one 1 in germination branch.
- Dedicated concentrate/backwash interfaces, CIP, sampling, training and commissioning records.

The well pump terminates at the raw-water isolation point. Civil pad/canopy/drainage and external electrical feeder remain in their own packages. The WTP supplier owns all internal skid piping, instruments and controls.

# 3. Governing Information and Design Basis

| Item | Adopted basis | Design note |
|---|---|---|
| Water source | Groundwater well | No raw water may bypass treatment into the NFT reserve group. |
| Net permeate duty | 5.0 m3/h | Equivalent to 40 m3 in one 8-hour production shift. |
| Design RO recovery | 70% baseline | Approx. 7.15 m3/h feed and 2.15 m3/h concentrate at full duty. |
| Reserve storage | 4 x 10 m3 = 40 m3 | Opaque food-grade PE tanks in one row; unchilled treated water. |
| Germination branch | 1 x 1 m3 tank | Independent 1 in treated-water branch to the service greenhouse. |
| WTP coordination envelope | 7.0 x 4.0 m clear, 3.0 m minimum headroom | Includes skid, filters, chemicals, service aisles and lifting access. |
| Operating location | South side, outside the service greenhouse | Aligned with the southern utility zone and protected by a ventilated canopy. |

Capacity check: 5.0 m3/h x 8 h = 40 m3. At 70% recovery, feed flow is 7.15 m3/h and concentrate flow is approximately 2.15 m3/h.

# 4. Treatment Process and Performance Requirements

## 4.1 Process train

| Step | Process | Execution requirement |
|---|---|---|
| 01 | Raw-water isolation and sample point | 3 in HDPE PN10 line; pressure, flow and laboratory sampling interface. |
| 02 | Raw feed / booster pump | Supplier-selected for minimum RO feed pressure at 7.2 m3/h after pipe losses. |
| 03 | Coarse pre-filtration | Hydrocyclone/disc or basket pre-filter to protect automatic valves and media. |
| 04 | Multimedia filtration | Turbidity and suspended-solids reduction with automatic backwash. |
| 05 | Iron/manganese treatment | Baseline vessel and bypass included; oxidation/media selected from verified well analysis. |
| 06 | Chemical conditioning | Chlorination/oxidation where required, antiscalant and membrane-compatible dechlorination/pH control. |
| 07 | 5 micron cartridge filtration | Final particulate barrier upstream of the high-pressure pump. |
| 08 | High-pressure RO skid | Membrane vessels, pump, valves, flow control, permeate and concentrate manifolds. |
| 09 | UV disinfection | SS316L chamber, 5 m3/h, minimum validated UV dose 40 mJ/cm2 at end of lamp life. |
| 10 | Treated-water distribution | 2 in product header to reserve tanks plus independent 1 in germination branch. |

## 4.2 Treated-water acceptance limits

| Parameter | Project limit | Verification |
|---|---|---|
| Feed turbidity to RO | <= 1.0 NTU | After pretreatment; stable under normal well operation. |
| SDI15 to RO | <= 3 | ASTM D4189 test or approved equivalent. |
| Permeate conductivity | <= 0.20 mS/cm at 25 degC | Or supplier-guaranteed value from the accepted membrane projection, whichever is stricter. |
| Permeate TDS | <= 150 mg/L | Target make-up water suitable for hydroponic nutrient formulation. |
| Total hardness | <= 50 mg/L as CaCO3 | Supports stable fertilizer mixing and limits precipitation. |
| Alkalinity | <= 60 mg/L as CaCO3 | Reduces acid demand and pH drift. |
| Iron / manganese | Fe <= 0.10; Mn <= 0.05 mg/L | Protects emitters, roots, tanks and UV transmission. |
| Salt rejection | >= 95% | Calculated from simultaneous feed and permeate conductivity readings. |
| UV dose | >= 40 mJ/cm2 | At design flow and end-of-lamp-life condition. |

**Hold point:** accredited well-water analysis and supplier membrane projection are accepted before final media, membrane array, chemical dose and guaranteed recovery release.

# 5. Equipment and Construction Requirements

| Code | Equipment | Qty | Minimum technical requirement |
|---|---|---|---|
| WTP-001 | Complete RO treatment station | 1 set | 5 m3/h net permeate; tropicalized skid and PLC panel. |
| WTP-002 | Raw feed/booster pump | 1 duty | 7.2 m3/h minimum at supplier-calculated TDH; dry-run protected. |
| WTP-003 | Coarse pre-filter | 1 set | Cleanable disc/basket or hydrocyclone arrangement. |
| WTP-004 | Multimedia filter | 1 set | Automatic service/backwash valves and differential-pressure indication. |
| WTP-005 | Iron/manganese filter | 1 set | Installed baseline vessel with bypass; final media by water analysis. |
| WTP-006 | Antiscalant dosing set | 1 set | 100 L minimum PE tank, duty dosing pump, mixer and low-level switch. |
| WTP-007 | 5 micron cartridge filter | 1 set | Multi-cartridge housing sized for 7.2 m3/h with pressure taps. |
| WTP-008 | RO high-pressure pump/membranes | 1 skid | Minimum 95% salt rejection at adopted feed-water design conditions. |
| WTP-009 | Instrumentation and PLC | 1 set | Pressure, flow, conductivity, alarms, flush and tank-level interlocks. |
| WTP-014 | Inline UV sterilizer | 1 set | SS316L, 5 m3/h, lamp-hour counter and UV-intensity alarm. |
| WTP-015 | Manual RO cleaning/CIP set | 1 set | PE cleaning tank, circulation pump, heater provision, hoses and 5 micron return filter. |
| WTP-016 | Chemical safety set | 1 set | Bunds, eyewash, SDS station, dosing calibration column and spill kit. |

- Downstream metallic water-contact parts are SS316L; carbon steel is prohibited in permeate contact.
- Skids are hot-dip galvanized or epoxy-coated; chemical tanks are opaque PE with bunds and lockable lids.
- Maintain 900 mm service clearance and full membrane-withdrawal/lifting access.
- Use a level concrete plinth, chemical-resistant finish, 1% floor fall and air-gapped drainage.

# 6. External Piping, Valves and Interfaces

| Code | Route | Pipe | CL | Buy | Requirement |
|---|---|---|---|---|---|
| WTR-01 | Well to WTP raw-water line | 3 in HDPE PN10 | 20 m | 21 m | Buried pressure line; final 3 x 2 in reducer at WTP inlet. |
| WTR-02A | WTP to reserve-room external product line | 2 in uPVC PN16 | 25 m | 27 m | Isolation, check valve, flow meter, EC/TDS and sample point. |
| WTR-02B | Reserve-room 2 in fill header | 2 in uPVC PN16 | 20 m | 23 m | Combined product route becomes 45 m centreline / 50 m procurement. |
| WTR-03 | Four reserve-tank top-fill branches | 1.5 in uPVC PN16 | 12 m | 14 m | One union, full-bore valve and flexible tank connector per branch. |
| WTR-04 | Germination treated-water branch | 1 in uPVC/PPR | 45 m | 48 m | Top-fill to 1 m3 tank; independent isolation and check valve. |
| WTR-05 | RO concentrate line | 2 in HDPE/uPVC PN10 | 45 m | 50 m | Continuous concentrate to lined evaporation/disposal route. |
| WTR-06 | Filter backwash connection | 3 in uPVC drainage | To WTP pad edge | Supplier set | Air-gapped discharge; external route belongs to site drainage. |
| WTR-07 | CIP drain/neutralization hose | 1.5 in chemical hose/uPVC | Local | Supplier set | Controlled batch discharge after pH verification. |

### 6.1 Standard valve and instrument sets

- Raw inlet: isolation, check valve, removable strainer, pressure gauge, flow meter and sample cock.
- Product outlet: isolation, check valve, totalizing flow, conductivity/temperature, UV, diversion valve and sample cock.
- Chemical pump: foot valve, calibration column, injection quill, back-pressure/anti-siphon, relief return and unions.
- Buried pipes: surveyed alignment, bedding, restrained changes, detectable tape and pressure testing before backfill.

## 7. Installation Sequence

- Release the pad, canopy, drainage, earth, sleeves and 7.0 x 4.0 m service envelope.
- Receive, verify and preserve skid, membranes, vessels, media, UV, instruments and chemicals.
- Install equipment without pipe strain and maintain service/membrane-removal clearances.
- Install and test raw, product and concentrate routes; cap and disinfect product piping.
- Load media, cartridges and membranes only after clean-water flushing and supplier inspection.
- Complete wiring, earthing, tank interlocks and diversion logic; calibrate all instruments.
- Commission pretreatment, RO, UV, product header and reserve filling in controlled stages.

## 8. Control Philosophy and Chemical Management

| Condition | Automatic action |
|---|---|
| Low raw-water pressure | Stop booster/high-pressure pump; alarm; no automatic restart until pressure is stable. |
| High cartridge differential pressure | Alarm and controlled shutdown if the supplier limit is exceeded. |
| High permeate conductivity | Divert to drain/recirculation; close reserve fill; alarm and operator acknowledgement. |
| Reserve tanks high-high | Close all product fill valves and stop WTP after automatic low-pressure flush. |
| Reserve group low level | Enable WTP automatic start, subject to raw-water availability and chemical levels. |
| Antiscalant low level / dosing fault | Prevent high-pressure pump start and issue visible/audible alarm. |
| UV lamp or intensity fault | Close product outlet or divert water; no untreated product to storage. |
| Membrane high pressure | Immediate high-pressure pump trip and latched alarm. |
| Emergency stop | De-energize pumps and chemical dosing; retain PLC alarm indication. |

- Bund capacity is 110% of the largest chemical container; incompatible chemicals use separate bunds.
- Only supplier-approved antiscalant and cleaning chemicals are used; every dosing pump is calibrated.
- CIP starts only at accepted normalized performance triggers; waste is neutralized before controlled discharge.

# 9. Testing, Commissioning and Acceptance

| Test | Method | Acceptance |
|---|---|---|
| Material verification | Check pipe ratings, vessel data, membrane serials and water-contact certificates. | Accepted material release and traceability register. |
| Pressure piping | Hydrotest separately at 1.5 times operating pressure or lower component limit. | No leakage or unexplained pressure loss. |
| Media filter | Service and backwash test with differential-pressure readings. | Uniform flow, no media carryover and correct valve sequence. |
| RO performance | Eight-hour continuous run at design conditions. | 5.0 m3/h net permeate, recovery and rejection within accepted projection. |
| Water quality | Accredited raw and permeate laboratory sampling plus online comparison. | All project water-quality limits achieved. |
| UV | Verify design flow, lamp hours, intensity alarm and shutdown/divert logic. | Minimum validated dose and correct interlock. |
| Tank interface | Simulate low, high and high-high levels for four reserve tanks. | Correct automatic start/stop and no overfill. |
| Reject/backwash | Operate concentrate and one filter backwash cycle. | Free discharge without surcharge, leakage or connection to treated water. |
| Integrated operation | Run WTP, reserve filling and germination branch under normal sequence. | Stable product quality, no cross-connection and complete records. |

## 9.1 Inspection and Test Plan

| No. | Inspection stage | Method | Type | Record |
|---|---|---|---|---|
| 01 | Approved well-water analysis and supplier projection | Document review | H | Design release |
| 02 | WTP pad, drainage, sleeves and service clearances | Survey/visual | H | Civil interface release |
| 03 | Skid, vessels, membranes, UV, chemicals and pipe materials | Identity/certificates | W | Material release |
| 04 | Raw/product/reject piping before concealment | Hydrotest/survey | H | Pressure test and as-built |
| 05 | Pretreatment media and automatic valve sequence | Functional test | W | Pretreatment report |
| 06 | RO pressure/leak/flush and membrane loading | Instrumented test | H | RO test pack |
| 07 | Water-quality and 8-hour performance test | Laboratory/performance | H | Commissioning report |
| 08 | Controls, alarms, diversion and tank interlocks | Cause-and-effect | H | Control acceptance |
| 09 | Training, spares, O&M and as-builts | Record review | R | Turnover dossier |

Intervention codes: H = Hold, W = Witness, S = Surveillance, R = Record review.

## 9.2 Hold points

- HP-WTR-01: Well analysis, process selection and projection accepted before procurement.
- HP-WTR-02: Pad, drainage, clearances and routes accepted before installation.
- HP-WTR-03: Pressure tests and route survey accepted before concealment.
- HP-WTR-04: Media, cartridges and membranes accepted before wet commissioning.
- HP-WTR-05: Diversion, UV and tank interlocks accepted before storage filling.
- HP-WTR-06: Eight-hour performance, laboratory results and records accepted before handover.

# 10. HSE, Environmental Controls and Handover

| Hazard | Mandatory control | Risk |
|---|---|---|
| Chemicals | SDS, labeled bunds, eyewash, PPE and controlled transfer | Medium |
| High pressure | Rated components, guards and controlled depressurization | Medium |
| Wet electrical | IP-rated equipment, earthing and LOTO | Medium |
| Media loading | Mechanical lifting, stable access and dust PPE | Medium |
| UV | Sealed interlocked chamber and isolation before service | Low |
| Wastewater | Approved lined route; no discharge to soil or treated tanks | Low |

Handover includes the accepted process flow, membrane projection, equipment curves, material/serial records, calibration and pressure tests, laboratory reports, normalized performance baseline, control cause-and-effect, as-builts, O&M manuals, warranties, training and critical spares.